

Madhu Varshini K

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Summary

Software Engineering student with practical experience in developing backend services, REST APIs, full-stack web applications, Android applications, and IoT solutions. Skilled in Java, Python, SQL, FastAPI, React, PostgreSQL, and Firebase. Strong understanding of Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, and Operating Systems. Quick learner with strong problem-solving skills and a willingness to adapt to new technologies.

Education

Integrated M.Tech in Software Engineering

Vellore Institute of Technology, Vellore

Expected 2027

CGPA: 8.7/10

Technical Skills

Programming Languages: Java, Python, C, C++, SQL

Web Technologies: HTML, CSS, JavaScript, React

Backend Technologies: FastAPI, REST APIs, JWT Authentication, Swagger (OpenAPI)

Databases: PostgreSQL, Firebase Realtime Database

Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

Tools & Platforms: Git, GitHub, Postman, Android Studio, Visual Studio Code

Projects

Effortless Grocery Shop | *React, FastAPI, PostgreSQL, JWT, REST APIs*

- Developed a full-stack grocery e-commerce application using React, FastAPI, and PostgreSQL.
- Designed and implemented REST APIs for user authentication, product management, cart operations, and order processing using JWT-based authentication.
- Built a recipe recommendation feature that enables users to create recipes and automatically add the required ingredients to the shopping cart.
- Integrated PostgreSQL with SQLAlchemy for efficient data storage and CRUD operations.

Personal Memory Assistant | *Python, FastAPI, PostgreSQL, pgvector, REST APIs*

- Developed an AI-powered memory assistant that stores and retrieves user memories using semantic search.
- Designed and implemented REST APIs with FastAPI for memory creation, retrieval, and semantic search.
- Generated vector embeddings for user memories using Sentence Transformers and stored them in PostgreSQL.
- Implemented semantic similarity search using pgvector to retrieve contextually relevant memories based on user queries.

Smart Helmet Safety & Emergency Alert System | *ESP32, Firebase, Bluetooth, Android*

- Developed an IoT-based smart helmet system for real-time accident detection and emergency alert generation.
- Built sensor data ingestion workflows using Firebase and integrated Bluetooth-based crash notifications.
- Implemented location tracking, emergency notifications, and real-time monitoring through Android integration.
- Designed a reliable safety solution to reduce emergency response time after accidents.

Publications & Patents

- **Research Paper:** *A Real-Time Crime Prediction and Resource Allocation Framework Using Machine Learning and Geospatial Analysis*, International Conference, NIT Puducherry, Dec 2025.
- **Indian Patent:** *IoT-Enabled Multi-State Medication Dispensing System with Sensor-Gated Dose Verification*, Published, 2026.

Achievements

- Received the **Best Paper Award** at **IconSCEPT 2025, NIT Puducherry** for the research paper “*A Real-Time Crime Prediction and Resource Allocation Framework Using Machine Learning and Geospatial Analysis*”.